MCA 2nd Semester External Lab Exam,2022

Sub: Data And File Structure Paper code:201\

Set –C

1. Write a program to delete a node from the end of a single linked list. Also display it.

2. Write a program to implement the features of a stack.

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MCA 2nd sem End Sem Practical Examination,July,2022

Sub: Graph Theory

Marks: 30 **Time: 1.5 hrs**

Set B

Q.1) Write a C Program to find the Intersection of two sets of integer elements.

Q.2) Write a C program to print the adjacency matrix of an undirected connected graph.

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MCA 2ND SEMESTER INTERNAL EXAMINATION

TIME :2HRS TOTAL MARKS:40

Answer the following:

Q1.Discuss the different components of data communication system . Explain different types of network topologies. 3+7=10

Q2.Explain the functions performed by different layers of the TCP/IP network model. 7

Q3.Discuss the different causes of transmission impairments. 5

Q4.What do you mean by Multiplexing in computer network ? Discuss different types of Multiplexing techniques. B 1+8=9

Q5.Write down the differences between serial and parallel transmission mode. 3

Q6.What do you mean by transmission media ? Mention its types. 1+2=3

Q7.Explain the importance of Shannon capacity and Nyquist data rate in communication. 3

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MCA 2nd Semester Internal Examination 2022

Sub: Object Oriented Programming and Design(Java)

tal Marks: 40

Time: 2 hrs.

Vhat is OOP ? Write the features of OOP .

2 + 5 = 7

xplain about the execution process of java.

5

Vhat are class and object ? Write a program in java by creating a class named 'Rectangle' with integer variables ngth' and 'width'. Assign the values of length and width as 20 and 10 respectively by creating an object of the class ctangle.

3 + 5 = 8

Vhat is Inheritance ? Explain about its different types.

2 + 6 = 8

Vrite about method overloading . Write a program in java by creating a class to print the result of multiplication of ger numbers. The class has two methods with the same name but different number of parameters.

3 + 4 = 7

efine Constructor. Write about its different types in java.

2 + 3 = 5

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12. Explain the phenomenon, how light produces on a phosphor coated screen.

display on a phosphor-coated screen.

4

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INTERNAL EXAMINATION – I

Sub: Computer Graphics (404) Marks: 40 Time: 1 Hr. MCA 2nd Semester Session: Jan – June, 22

1. Define Computer graphics. 1
2. Define Random and Raster scan displays? 2
3. What is resolution? 1
4. What is a frame buffer? 1
5. Explain how colour images can display on a thin-film electroluminescent display. 3
6. What do you mean by emissive and non-emissive displays? 2
7. Explain the characteristics of polarized light. 2
8. Define the term liquid crystal? 2

9. Given the centre point coordinates (10, 10) and radius as 10, generate all the points to form a circle. 6

10. Calculate the points between the starting coordinates (9, 18) and ending coordinates (14, 22) using Bresenham's line drawing algorithm. 6
11. Draw a line segment using Bresenham's line drawing line drawing algorithm with the starting coordinate (10, 15) and the endpoint coordinate (20, 25). 6
12. Explain the phenomenon, how light produced on a phosphor-coated screen. 4
13. Explain how colour images can display on a phosphor-coated screen. 4

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Total number of printed pages–4

2 SEM MCA (CBCS) DFS 1

2022

(July)

COMPUTER APPLICATION

Paper : 201

(Data and File Structures)

Full Marks : 60

Time : Three hours

The figures in the margin indicate full marks for the questions.

Question number **1** *is* ***Compulsory.***

Attempt ***any four*** *questions from the rest.*

1. Answer the following : 2×10=20

   (a) What is a balanced tree and what is its importance ?

   (b) How is circular queue represented ?

   (c) What is the maximum number of nodes in a binary tree of height k ?

Contd.

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(d) Mention *two* important properties of binary tree.

(e) What is indexed sequential file ?

(f) What is a strict binary tree ?

(g) What is a heap ?

(h) Calculate the efficiency of sequential search.

(i) Which data structures are applied when dealing with a recursive function ?

(j) Differentiate between NULL and VOID.

2. (a) Write binary search algorithm. Find its time complexity. 5

(b) The following sequence lists the nodes of a binary tree T in preorder and inorder respectively ;

Preorder : G, B, Q, A, C, K, F, P, D, E, R, H.

Inorder : Q, B, K, C, F, A, G, P, E, D, H, R

Draw the diagram of the tree. 5

3. (a) Show the stages of growth of an order-4 B-tree when the following keys are inserted in the order given : 6

84, 82, 29, 99, 65, 12, 50, 28, 58, 71, 92, 75, 92, 75, 19, 55

(b) Construct an AVL tree from the following numbers : 4

63, 9, 19, 27, 18, 108, 99, 81

4. (a) Compare between quick sort and selection sort in terms of time and space complexities. 5

(b) Explain the concept of multilist with an example. 5

5. (a) Find the complexity for the following codes : 4

```
int count = 0;
for (int i = 0; i<N; i++)
    for (intj = 0; j<i; j++)
        count++;
```

(b) What is sequential file organization ? What are its disadvantages ? Explain how indexed sequential file organization is better than sequential file organization. 6

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6. *(a)* Build a max heap H from the given set of numbers :

45, 36, 54, 27, 63, 72, 61, 18

From the heap H delete 72. 6

(b) Write a recursive function to perform selection sort. 4

Preorder -
Inorder - parent left-right
Inorder - left - parent right
Postorder - left-right-parent

(63)

(63)

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2 SEM MCA (CBCS) DCCN 2

2022

(July)

COMPUTER APPLICATION

Paper : 202

(Data Communications and Computer Networks)

Full Marks : 60

Pass Marks : 24

Time : Three hours

The figures in the margin indicate full marks for the questions.

1. *(a)* How are OSI and ISO related to each other ? 1

(b) Match the following to one or more layers of OSI model : 4

(i) Communicates directly with user's application program Application

(ii) Error correction and retransmission Transport

(iii) Mechanical, electrical and functional interface

(iv) Responsibility for carrying frames between adjacent nodes

Contd.

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(c) Compare and contrast the functionalities of transport layer and data link layer of OSI model. 5

2. (a) Define framing and the reason for its need. Compare between bit stuffing and byte stuffing. 1+4=5

(b) Briefly explain the stop and wait protocol for noiseless channels. 5

OR

Briefly explain the stop and wait ARQ for noisy channels. 5

3. (a) Compare and contrast symmetric key cryptography and asymmetric key cryptography. 5

(b) Describe shift cipher with a suitable example. 3

OR

Describe transposition cipher with a suitable example. 3

(c) What are different types of P-boxes ? 2

4. (a) What is ALOHA ? Briefly explain the procedure for pure ALOHA. 1+5=6

OR

Briefly explain different persistent methods. 6

(b) How does CSMA/CA differ from CSMA/CD ? 4

5. (a) What is UDP ? Define user datagram and explain the user datagram format. 1+5=6

OR

Describe the connection oriented process of TCP with suitable diagram. 6

(b) How does TCP differ from UDP ? 4

6. (a) Briefly explain the working principle of distance vector routing protocol. 8

OR

Describe the working principles of link state routing protocol. 8

(b) What is BGP ? 2

Weights, expansion and compression

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Total number of printed pages–3

2 SEM MCA (CBCS) OOPD 4

2022

(July)

COMPUTER APPLICATION

Paper : 204

[Object Oriented Programming and Design (Java)]

Full Marks : 60

Pass Marks : 27

Time : Three hours

The figures in the margin indicate full marks for the questions.

1. Answer ***any three*** from the following :

(a) (i) Why can't we use a keyword as a variable name ? 2

(ii) When do we declare a member of a class static ? 2

(b) What is class ? How does it accomplish data hiding ? 2+2=4

(c) Describe the structure of a typical Java program. 4

Contd.

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(d) Describe the process of converting bytecode into machine code. Draw the necessary diagram. 4

2. Answer **any two** from the following :

(a) Describe different forms of inheritance with examples. 6

(b) Describe the similarities between interfaces and classes. 6

(c) Describe method overriding. Give example. 6

3. Answer the following :

(a) How do we set priorities for thread ? 2

(b) Differentiate between suspending and stopping a thread. 3

(c) (i) What is finally block ? When and how is it used ? Give a suitable example. 1+2+4=7

Or

(ii) Create a try block that is likely to generate three types of exception and then incorporate necessary catch blocks to catch and handle them appropriately. 7

4. Answer the following : 2

(a) Define collection.

(b) (i) Write a program, which will read a string and rewrite it in the alphabetical order. For example, the word STRING should be written as GINRST. 10

Or

(ii) Write a program, which will read a text and count all occurrences of a particular word.

5. Answer the following :

(a) Differentiate between aggregation and composition. 4

(b) (i) What is activity diagram ? Design an activity diagram for online examination system. 2+6=8

Or

(ii) Describe the class relationships. Give example. 8

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Total number of printed pages–6

2 SEM MCA (CBCS) GTA 2

2022

(July)

COMPUTER APPLICATION

Paper : 206

(Graph Theoretical Algorithms)

Full Marks : 60

Time : Three hours

The figures in the margin indicate full marks for the questions.

1. Define the following terms with example : 2×5=10

(a) Path

(b) Regular graph

(c) Unicursal line

(d) Chromatic number

(e) Complete graph

Contd.

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2. Answer **any five** of the following questions : 3×5=15

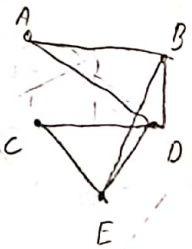
(a) A simple graph contains 35 edges, four vertices of degree 5, five vertices of degree 4 and four vertices of degree 3. Find the number of vertices with degree 2.

(b) What is bipartite graph ? Explain with an example.

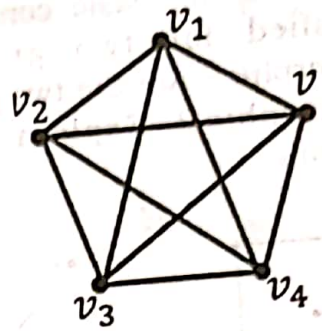
(c) Define Hamiltonian graph. Find an example of a non-Hamiltonian graph with a Hamiltonian path.

(d) Draw the graph for the following adjacency matrix :

$$X(G) = \begin{array}{c} \\ A\ 1 \\ B\ 2 \\ C\ 3 \\ D\ 4 \\ E\ 5 \end{array} \begin{array}{c} \begin{array}{ccccc} 1\ A & 2\ B & 3\ C & 4\ D & 5\ E \end{array} \\ \begin{bmatrix} 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 & 0 \end{bmatrix} \end{array}$$



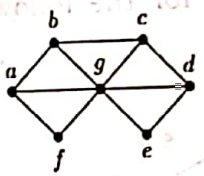
(e) Check whether the following graph is planar or not :



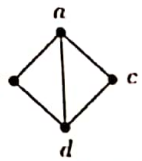
(f) Find the chromatic polynomial of the following graphs :

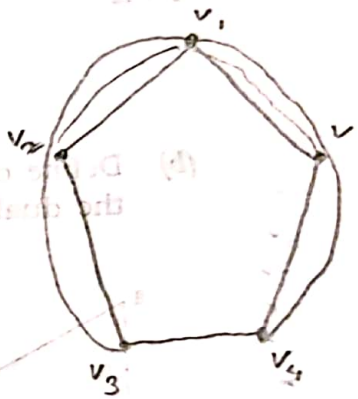
A =

B =



C =





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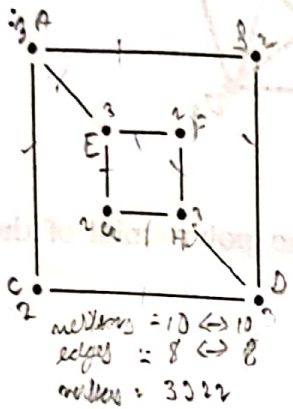
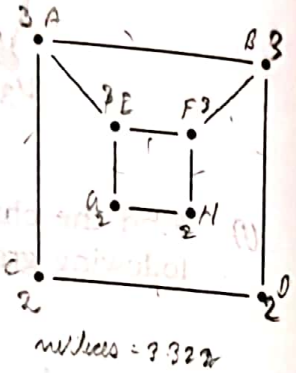
3. Answer ***any five*** from the following questions : 5×5=25

(a) What are the basic conditions to be satisfied for two graphs to be isomorphic? Are the two graphs below isomorphic? Explain with valid reasons.

G :



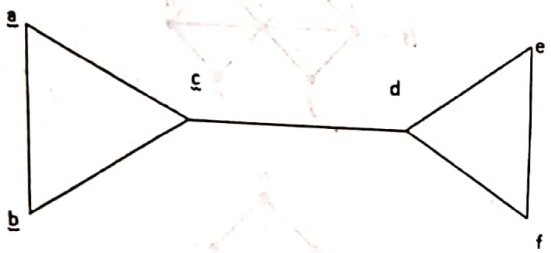
H :



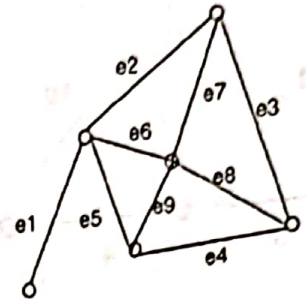
(b) Define dual of a planar graph. Draw the dual for the following graph :

A-A'
B-B'
C-C'
D-D'
E-E'
F-F'
G-G'
H-H'

A
B
C
D
E
F
G
H

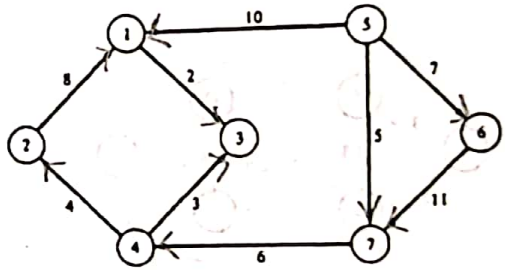


(c) Define cut set. Find *any four* cut sets from the graph *G* given below and also find the edge connectivity of *G*.



(d) What is Euler graph? Prove the Eulerian necessary and sufficient condition.

(e) Find the incidence matrix of the following directed graph :



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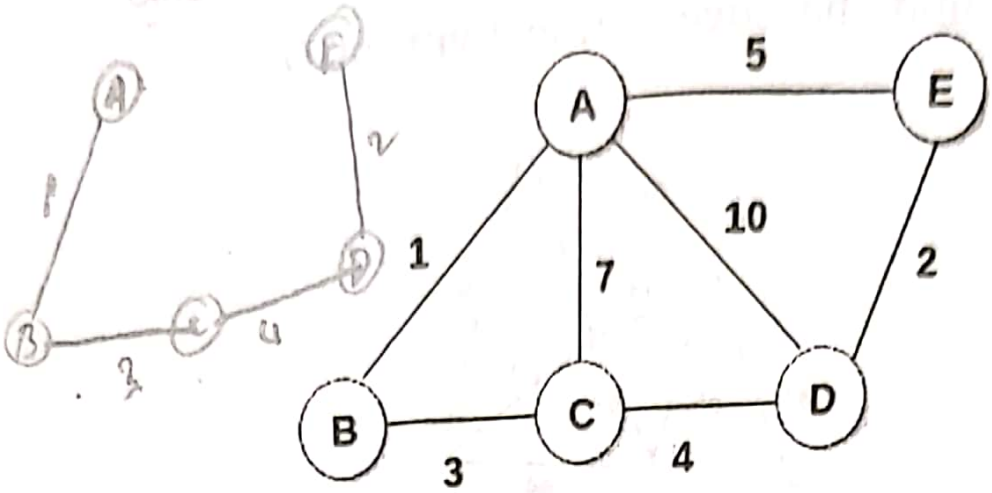
(1) Find MST from the graph given below using Kruskal's algorithm :

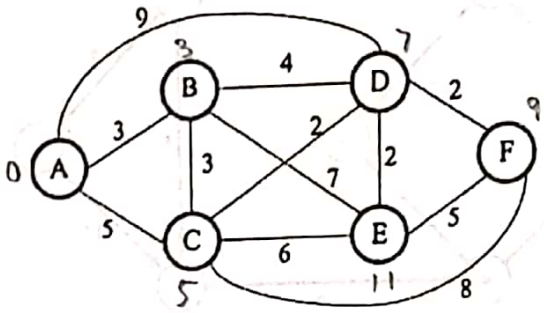


4. Prove that chromatic polynomial of complete graph is

$$P_n(\lambda) = \lambda(\lambda-1)(\lambda-2).....(\lambda-n+1)$$ 10

Or

Find single source shortest path using Dijkstra's algorithm from the following graph :



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2 SEM MCA (CBCS) CGM 8

2022

(July)

COMPUTER APPLICATION

Paper : 208

(Computer Graphics and Multimedia)

Full Marks : 60

Time : Three hours

The figures in the margin indicate full marks for the questions.

1. Answer the following : ***(any five)*** 2×5=10

   (a) What is aspect ratio of a CRT ?

   (b) Write down the importance of homogeneous coordinates.

   (c) State the concept of vanishing point.

   (d) What is rendering ?

Contd.

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(e) Prove that successive two dimensional rotations are additive.

(f) What are the differences between raster image and vector image ?

2. Write down Bresenham's line drawing algorithm. Explain with an example. 6

3. Explain about the steps involved in 2D viewing pipeline system with proper diagram. 6

4. Explain different types of animation. What are the fundamental principles of animation ? 4+4=8

5. What is video controller ? Explain the working of a video controller in a CRT. 2+4=6

6. Explain with diagram about reflection and rotation in 2D transformation. 3+3=6

7. Explain perspective projection with its types. 6

8. Use Cohen-Sutherland algorithm to clip line P_1 (70, 20) and P_2 (100, 10) against the window lower left-hand corner (50, 10) and upper right-hand corner (80, 40). 6

OR

Consider we have a square $O(0, 0)$, $B(4, 0)$, $C(4, 4)$, $D(0, 4)$ on which we first apply T_1 (scaling transformation) given scaling factor is $Sx = Sy = 0.5$ and then we apply T_2 (rotation transformation in clockwise direction) by 90* (angle), and at last we perform T_3 (reflection transformation about origin). 6

9. Write short notes on ***any two*** of the following : 3×2=6

(a) Image restoration

(b) Homogeneous coordinates

(c) GUI

P_1 (9, 8 4) Bottom left

P_2 (

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